MCE5919: Gaussian Process for Machine Learning and Signal Processing

Lecture 3

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Overview

1. Bayesian Linear Regression

2. Kernel and Kernel Tricks

3. Kernelized Bayesian Regression: Primal vs Dual

4. Appendix

A Gentle Begin

- Given a dataset $\mathcal{D} = \{y_i, \mathbf{x}_i\}_{i=1}^N$ with $y_i \in \mathbb{R}$ and $\mathbf{x}_i \in \mathbb{R}^K$
- Denote a parameterized linear model $f(\mathbf{x}; \boldsymbol{\theta})$ as

$$f(\mathbf{x}; \boldsymbol{\theta}) = \sum_{k=1}^{K} \theta_i x_i$$

  - model parameter – $\boldsymbol{\theta} = (\theta_0, \theta_1, \ldots, \theta_K) \in \mathbb{R}^K$
  - input/feature vector – $\mathbf{x} = (x_0, x_1, \ldots, x_K) \in \mathbb{R}^K$
- **Data Fitting View:** Find $\boldsymbol{\theta}$ that 'best' fitting the dataset $\mathcal{D}$

$$\texttt{for each } i, \quad y_i - f(\mathbf{x}_i; \boldsymbol{\theta}) = \varepsilon_i \quad \rightarrow \quad \texttt{minimize } \sum_i \|\varepsilon_i\|^2$$

Remark

- ε_i represents model mismatch
- $\|\cdot\|$ is the Euclidean norm and can be replaced by other metrics

Linear Regression

- Compactly the data model becomes

$$\mathbf{y} = \mathbf{X}^T\boldsymbol{\theta} + \boldsymbol{\varepsilon}$$

  - $\mathbf{y} = (y_1, y_2, \ldots, y_N) \in \mathbb{R}^N$, $\mathbf{X} = (\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_N)^T \in \mathbf{R}^{N\times K}$, $\boldsymbol{\varepsilon} = (\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_N) \in \mathbb{R}^N$
- Finding $\boldsymbol{\theta}$ such that minimizing $\|\boldsymbol{\varepsilon}\|^2$ is formulated as

$$\min_{\boldsymbol{\theta}} \ \|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}\|^2 \tag{1}$$

- Supposing $\mathbf{X}$ is full-column rank ($N \geq K$), Problem (1) is strongly convex
- Its unique optimal solution can be attained in an analytic form

$$\boldsymbol{\theta}^* = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$$

Remark:

- $\boldsymbol{\theta}^*$ is deterministic w.r.t. dataset $\mathcal{D} = \{\mathbf{y}, \mathbf{X}\}$, no assumptions on $\boldsymbol{\varepsilon}$
- If $\varepsilon_i \sim \mathcal{N}(0, \sigma^2), \forall i$, $\boldsymbol{\theta}^*$ is also known as the maximum likelihood estimator

$$\boldsymbol{\theta}^* = \arg\max_{\boldsymbol{\theta}} p(\mathbf{y} \mid \mathbf{X}, \boldsymbol{\theta})$$

Bayesian Setting

- Suppose we have additional prior information on $\boldsymbol{\theta}$, e.g., $p(\boldsymbol{\theta}) \sim \mathcal{N}(\mathbf{0}, \mathbf{C})$

Limitation of Deterministic Setting

- Useful prior can not be incorporated under deterministic setting

Bayesian Setting

- Goal: Making use of the prior $p(\boldsymbol{\theta})$!
- Two tasks in Bayesian linear regression (find posterior distribution)
  - Given $\mathcal{D} = \{\mathbf{y}, \mathbf{X}\}$, find out the posteriror distribution of $\boldsymbol{\theta}$

$$p(\boldsymbol{\theta} \mid \mathbf{y}, \mathbf{X}) = ?$$

  - Given new input/feature $\mathbf{x}'$, find out the posteriror distribution of prediction y'

$$p(y' \mid \mathbf{x}', \mathbf{y}, \mathbf{X}) = ?$$

Bayes Theorem

- **Core Idea:** Bayes Theorem
  - $p(\boldsymbol{\theta})$ – the prior distribution
  - $p(\mathbf{y} \mid \mathbf{X}, \boldsymbol{\theta})$ – the likelihood given $\boldsymbol{\theta}$

$$p(\boldsymbol{\theta} \mid \mathbf{y}, \mathbf{X}) = \frac{p(\mathbf{y} \mid \mathbf{X}, \boldsymbol{\theta}) \cdot p(\boldsymbol{\theta})}{p(\mathbf{y} \mid \mathbf{X})}$$

  - $p(\mathbf{y} \mid \mathbf{X})$ – normalized constant, also known as marginal likelihood

$$p(\mathbf{y} \mid \mathbf{X}) = \int p(\mathbf{y} \mid \mathbf{X}, \boldsymbol{\theta}) \cdot p(\boldsymbol{\theta}) d\boldsymbol{\theta}$$

Remark:

- The posterior absorbs knowledge from both the likelihood and the prior

- Next let do some math for figuring out

$$p(\boldsymbol{\theta} \mid \mathbf{y}, \mathbf{X}) = ? \quad p(y' \mid \mathbf{x}', \mathbf{y}, \mathbf{X}) = ?$$

Find $p(\boldsymbol{\theta} \mid \mathbf{y}, \mathbf{X})$

How to find $p(\boldsymbol{\theta} \mid \mathbf{y}, \mathbf{X}) = ?$

- Recall linear model

$$y_i = f(\mathbf{x}_i; \boldsymbol{\theta}) + \varepsilon_i = \mathbf{x}_i^T \boldsymbol{\theta} + \varepsilon_i, \ \forall i$$

- Recall Gaussian assumptions

$$\varepsilon_i \sim \mathcal{N}(0, \sigma^2), \quad p(\boldsymbol{\theta}) \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$$

- Both the prior $p(\boldsymbol{\theta})$ and likelihood $p(\mathbf{y} \mid \mathbf{X}, \boldsymbol{\theta})$ can be explicitly expressed

Find $p(\boldsymbol{\theta} \mid \mathbf{y}, \mathbf{X})$

- The likelihood function $p(\mathbf{y} \mid \mathbf{X}, \boldsymbol{\theta})$ is

$$
\begin{aligned}
p(\mathbf{y} \mid \mathbf{X}, \boldsymbol{\theta}) &= \prod_{i=1}^{n} p\left(y_i \mid \mathbf{x}_i, \boldsymbol{\theta}\right) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{\left(y_i - \mathbf{x}_i^T \boldsymbol{\theta}\right)^2}{2\sigma^2}\right) \\
&= \frac{1}{\left(2\pi\sigma^2\right)^{n/2}} \exp\left(-\frac{1}{2\sigma^2} \left\| \mathbf{y} - \mathbf{X}^T \boldsymbol{\theta} \right\|^2\right) \\
&= \mathcal{N}\left(\mathbf{X}^T \boldsymbol{\theta}, \sigma^2 \mathbf{I}\right)
\end{aligned}
$$

- The prior function $p(\boldsymbol{\theta})$

$$
p(\boldsymbol{\theta}) = \mathcal{N}\left(\mathbf{0}, \boldsymbol{\Sigma}\right) = \frac{1}{(2\pi)^{N/2} \sqrt{|\boldsymbol{\Sigma}|}} \exp\left[-\frac{1}{2} \boldsymbol{\theta}^T \boldsymbol{\Sigma}^{-1} \boldsymbol{\theta}\right]
$$

Find $p(\theta \mid \mathbf{y}, \mathbf{X})$

- Derivation for the posterior $p(\boldsymbol{\theta} \mid \mathbf{y}, \mathbf{X})$

$$
\begin{aligned}
p(\boldsymbol{\theta} \mid \mathbf{X}, \mathbf{y}) &= \frac{p(\mathbf{y} \mid \mathbf{X}, \boldsymbol{\theta}) p(\boldsymbol{\theta})}{p(\mathbf{y} \mid \mathbf{X})} \\
&\propto p(\mathbf{y} \mid \mathbf{X}, \boldsymbol{\theta}) p(\boldsymbol{\theta}) \\
&\propto \exp\left[-\frac{1}{2\sigma^2}\left\|\mathbf{y} - \mathbf{X}^T\boldsymbol{\theta}\right\|^2\right] \exp\left[-\frac{1}{2}\boldsymbol{\theta}^T\boldsymbol{\Sigma}^{-1}\boldsymbol{\theta}\right] \\
&= \exp\left[-\frac{1}{2}\boldsymbol{\theta}^T(\sigma^{-2}\mathbf{X}\mathbf{X}^T + \boldsymbol{\Sigma}^{-1})\boldsymbol{\theta} + \sigma^{-2}(\mathbf{X}\mathbf{y})^T\boldsymbol{\theta} - \frac{1}{2}\sigma^{-2}\|\mathbf{y}\|^2\right]
\end{aligned}
$$

Find $p(\boldsymbol{\theta} \mid \mathbf{y}, \mathbf{X})$

How to simplify $\frac{1}{2}\boldsymbol{\theta}^T(\sigma^{-2}\mathbf{X}\mathbf{X}^T + \boldsymbol{\Sigma}^{-1})\boldsymbol{\theta} + \sigma^{-2}(\mathbf{X}\mathbf{y})^T\boldsymbol{\theta} - \frac{1}{2}\sigma^{-2}\|\mathbf{y}\|^2$

- Using the fact $\frac{1}{2}(\boldsymbol{\theta} - \boldsymbol{\mu}_\theta)^T\mathbf{C}(\boldsymbol{\theta} - \boldsymbol{\mu}_\theta) = \frac{1}{2}\boldsymbol{\theta}^T\mathbf{C}\boldsymbol{\theta} - \boldsymbol{\mu}_\theta^T\mathbf{C}\boldsymbol{\theta} + \text{constant}$
- Specifying $\mathbf{C}$ and $\boldsymbol{\mu}_\theta$

$$\mathbf{C} = (\sigma^{-2}\mathbf{X}\mathbf{X}^T + \boldsymbol{\Sigma}^{-1}), \quad \boldsymbol{\mu}_\theta^T\mathbf{C} = \sigma^{-2}(\mathbf{X}\mathbf{y})^T \Rightarrow \boldsymbol{\mu}_\theta = \sigma^{-2}\mathbf{C}^{-1}(\mathbf{X}\mathbf{y})$$

- We can get

$$p(\boldsymbol{\theta} \mid \mathbf{X}, \mathbf{y}) \propto \exp\left[-\frac{1}{2}(\boldsymbol{\theta} - \boldsymbol{\mu}_\theta)^T\mathbf{C}(\boldsymbol{\theta} - \boldsymbol{\mu}_\theta)\right] = \mathcal{N}(\boldsymbol{\mu}_\theta, \mathbf{C}^{-1})$$

- **The posterior of θ is also a Gaussian distribution**

Find $p(y' \mid \mathbf{x}', \mathbf{y}, \mathbf{X})$

Given new input/feature $\mathbf{x}'$, how to find $p(y' \mid \mathbf{x}', \mathbf{y}, \mathbf{X}) = ?$

- **Computational View:** $\boldsymbol{\theta} \sim p(\boldsymbol{\theta} \mid \mathbf{X}, \mathbf{y})$, predict as $y' = \boldsymbol{\theta}^T \mathbf{x}' (+\varepsilon')$ (infinite many $\boldsymbol{\theta}$)
- **Statistical View:** $y' \sim \mathcal{N}(\mu_y, \sigma_y^2)$, marginalize the prediction of all possible $\boldsymbol{\theta}$

$$p(y' \mid \mathbf{x}', \mathbf{y}, \mathbf{X}) = \int p(y' \mid \mathbf{x}', \boldsymbol{\theta}) p(\boldsymbol{\theta} \mid \mathbf{X}, \mathbf{y}) d\boldsymbol{\theta}$$

- By the closure properties of Gaussians, $p(y' \mid \mathbf{x}', \mathbf{y}, \mathbf{X}) = \mathcal{N}(\mu_y, \sigma_y^2)$ still a Gaussian

$$\mu_y = \mathbb{E}\left[y'\right] = \boldsymbol{\mu}_\theta^T \mathbf{x}'$$
$$\sigma_y^2 = \mathbb{VAR}\left[y'\right] = (\mathbf{x}')^T \mathbf{C} \mathbf{x}' + \sigma^2$$

($\mathbf{C}$ and $\boldsymbol{\mu}_\theta$ are given before)

Short Summary

Summary:

- Given $\mathcal{D} = \{\mathbf{y}, \mathbf{X}\}$ and linear model $\mathbf{y} = \mathbf{X}^T\boldsymbol{\theta} + \varepsilon$
- Linear regression

$$\boldsymbol{\theta}^* = \arg\max_{\boldsymbol{\theta}} p(\mathbf{y} \mid \mathbf{X}, \boldsymbol{\theta}) = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$$

- Bayesian linear regression

$$p(\boldsymbol{\theta} \mid \mathbf{X}, \mathbf{y}) = \mathcal{N}(\boldsymbol{\mu}_\theta, \mathbf{C}^{-1}), \quad p(y' \mid \mathbf{x}', \mathbf{y}, \mathbf{X}) = \mathcal{N}(\mu_y, \sigma_y^2)$$

Question: How to deal nonlinear model? $\rightarrow$ Kernel Tricks

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Projection into New Feature Space

- Bayesian linear models suffer from limited expressiveness
- **Kernel Trick** (Implementation View)
  - project the inputs into some high dimensional space using a set of basis functions
  - apply linear model in high dimensional space
- **Example** (scalar input)
  - polynomial basis: $\phi(x) = (1, x, x^2, x^3, ...)$
  - trigonometric basis: $\phi(x) = (1, \cos \omega_1 x, \sin \omega_1 x, \cos \omega_2 x \dots)$
  - cubic–spline basis: $\phi(x) = (1, x, (x - c_1)_+^2, \dots)$
- Turns *linear* learner into powerful *non-linear* learners by hiding the feature map
- Feature spaces can be finite or infinite large — kernels unify them all
- Choosing the right kernel = choosing the right inductive bias for your data

Making Prediction in New Space

Recall: The predictive distribution for the linear model in the original input space is

$$p(y' \mid \mathbf{x}', \mathbf{y}, \mathbf{X}) = \mathcal{N}(\mu_y, \sigma_y^2) \tag{2}$$

where

$$\mu_y = \mathbb{E}\left[y'\right] = \boldsymbol{\mu}_\theta^T \mathbf{x}', \quad \sigma_y^2 = \mathbb{VAR}\left[y'\right] = (\mathbf{x}')^T \mathbf{C} \mathbf{x}' + \sigma^2$$
$$\mathbf{C} = (\sigma^{-2} \mathbf{X}\mathbf{X}^T + \boldsymbol{\Sigma}^{-1}), \quad \boldsymbol{\mu}_\theta = \sigma^{-2} \mathbf{C}^{-1} (\mathbf{X}\mathbf{y})$$

Lifting to High-Dimensional Space

- Introduce a mapping function $\phi(x) : \mathbb{R}^K \mapsto \mathbb{R}^P$
  - maps a K-dimensional vector into a new P-dimensional feature vector ($P \gg K$)
- Define a linear model in the new feature space (high-dimensional) as

$$f(\mathbf{x}) = \phi(\mathbf{x})^T \mathbf{w}$$

  - v.s. linear model $f(\mathbf{x}) = \mathbf{x}^T \boldsymbol{\theta}$
- Define matrix $\Phi(\mathbf{X})$ as the aggregation of columns $\phi(\mathbf{x})$

$$\Phi(\mathbf{X}) = [\phi(\mathbf{x}_1), \phi(\mathbf{x}_2), ..., \phi(\mathbf{x}_N)] \in \mathbb{R}^{P \times N}$$

- Concretely, data model becomes ($\varepsilon \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})$, $p(\mathbf{w}) \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$)

$$\mathbf{y} = \Phi(\mathbf{X})^T \mathbf{w} + \varepsilon$$

Predictive Distribution

- Use exactly the same derivation techniques!
- The predictive distribution in the new feature space is ($\mathbf{\Phi} := \Phi(\mathbf{X})$):

$$p(y' \mid \mathbf{x}', \mathbf{y}, \mathbf{X}) = \mathcal{N}(\mu_y, \sigma_y^2)$$

$$\mu_y = \mathbb{E}\left[y'\right] = \boldsymbol{\mu}_w^T \boldsymbol{\phi}(\mathbf{x}'), \quad \sigma_y^2 = \mathbb{VAR}\left[y'\right] = \boldsymbol{\phi}(\mathbf{x}')^T \mathbf{C} \boldsymbol{\phi}(\mathbf{x}') + \sigma^2$$
$$\mathbf{C} = (\sigma^{-2} \mathbf{\Phi}\mathbf{\Phi}^T + \mathbf{\Sigma}^{-1}), \quad \boldsymbol{\mu}_w = \sigma^{-2} \mathbf{C}^{-1} (\mathbf{\Phi}\mathbf{y})$$

- Computational complexity is prohibitive for large P!

Complexity Issue: Predictive Mean

The predictive mean in original form is

$$\mu_y = \sigma^{-2}\phi\left(\mathbf{x}'\right)^T \mathbf{C}^{-1}\mathbf{\Phi}\mathbf{y}, \quad \mathbf{C} = (\sigma^{-2}\mathbf{\Phi}\mathbf{\Phi}^T + \mathbf{\Sigma}^{-1})$$

Huge $\mathcal{O}(P^3)$ complexity $\Rightarrow$ $\mathcal{O}(N^3)$ complexity $(P \gg K)$

1. First, we introduce

$$\mathbf{C}\mathbf{\Sigma}\mathbf{\Phi} = (\sigma^{-2}\mathbf{\Phi}\mathbf{\Phi}^T + \mathbf{\Sigma}^{-1})\mathbf{\Sigma}\mathbf{\Phi} = \sigma^{-2}\mathbf{\Phi}\mathbf{\Phi}^T\mathbf{\Sigma}\mathbf{\Phi} + \mathbf{\Phi}$$

2. Then, multiply by $(\mathbf{\Phi}^T\mathbf{\Sigma}\mathbf{\Phi} + \sigma^2\mathbf{I})^{-1}$ from right and $\mathbf{C}^{-1}$ from left:

$$\mathbf{\Sigma}\mathbf{\Phi}(\mathbf{\Phi}^T\mathbf{\Sigma}\mathbf{\Phi} + \sigma^2\mathbf{I})^{-1} = \sigma^{-2}\mathbf{C}^{-1}\mathbf{\Phi}$$

3. Finally, substitute in the expression of the mean μ:

$$\mu_y = \sigma^{-2}\phi\left(\mathbf{x}'\right)^T \mathbf{C}^{-1}\mathbf{\Phi}\mathbf{y} = \phi\left(\mathbf{x}'\right)^T \mathbf{\Sigma}\mathbf{\Phi}(\mathbf{\Phi}^T\mathbf{\Sigma}\mathbf{\Phi} + \sigma^2\mathbf{I})^{-1}\mathbf{y}$$

Complexity Issue: Predictive Covariance

For the same reason, we reformulate the original predictive covariance

$$\sigma_y^2 = \phi\left(\mathbf{x}'\right)^T \mathbf{C}^{-1} \phi\left(\mathbf{x}'\right) + \sigma^2, \quad \mathbf{C} = (\sigma^{-2}\mathbf{\Phi}\mathbf{\Phi}^T + \mathbf{\Sigma}^{-1})$$

- Apply the matrix inversion lemma

$$(Z + UWV^T)^{-1} = Z^{-1} - Z^{-1}U(W^{-1} + V^T Z^{-1} U)^{-1} V^T Z^{-1}. \tag{3}$$

- Set $Z^{-1} = \mathbf{\Sigma}$, $W^{-1} = \sigma^2 \mathbf{I}$, $V = U = \mathbf{\Phi}$.
- Finally, we obtain

$$\sigma_y^2 = \phi(x')^T \mathbf{\Sigma} \phi(x') - \phi(x')^T \mathbf{\Sigma}\mathbf{\Phi} \left(\mathbf{\Phi}^T \mathbf{\Sigma}\mathbf{\Phi} + \sigma^2 \mathbf{I}\right)^{-1} \mathbf{\Phi}^T \mathbf{\Sigma} \phi(x') \tag{4}$$

First Exposure to Kernel

- Notice that $\phi(\mathbf{x}_1)^T \mathbf{\Sigma} \phi(\mathbf{x}_2)$ is essentially a dot product, why?
- With the aid of Cholesky decomposition: $\mathbf{\Sigma} = \mathbf{L}^T \mathbf{L}$, we define a kernel function:

$$k(\mathbf{x}_1, \mathbf{x}_2) = \psi(\mathbf{x}_1)^T \psi(\mathbf{x}_2), \tag{5}$$

  where we redefined $\psi(\mathbf{x}) := \mathbf{L}\phi(\mathbf{x})$
- **Important Observation**: the feature space ϕ appears always in the form

$$\phi(\mathbf{x}_1)^T \mathbf{\Sigma} \phi(\mathbf{x}_2)$$

- We can rely on the kernel instead of the mapping function

Kernel Tricks

- Feature mapping is often done from low-dimensional to high-dimensional spaces, i.e.,

$$\phi(\mathbf{x}) : \mathbb{R}^K \mapsto \mathbb{R}^P \ (P \gg K)$$

- For P being large, computing $\phi(\mathbf{x})^T \phi(\mathbf{x}')$ is unfeasible
- A computationally efficient way is adopting an equivalent kernel function

$$k(\mathbf{x}, \mathbf{x}') = (\approx)\phi(\mathbf{x})^T \phi(\mathbf{x}')$$

- In Gaussian process model, we work directly with kernel function.

Relationship Between ϕ and Kernel

- A kernel is an inner product in (possibly high-dimensional) feature space

$$k(\mathbf{x}, \mathbf{x}') := \langle \phi(\mathbf{x}), \phi(\mathbf{x}') \rangle = \phi(\mathbf{x})^T \phi(\mathbf{x}')$$

- **Kernel trick:** algorithms that depend on ϕ only through dot-products can be written using $k(\cdot, \cdot)$ without explicitly computing ϕ
- Examples
  - Linear kernel: $k(\mathbf{x}, \mathbf{x}') = \mathbf{x}^T \mathbf{x}'$ (take $\phi(\mathbf{x}) = \mathbf{x}$, finite-dimensional)
  - Polynomial kernel: $k(\mathbf{x}, \mathbf{x}') = (\mathbf{x}^T \mathbf{x}' + c)^d$ (equivalent to all monomials up to degree d, finite-dimensional)
  - RBF/Gaussian kernel: $k(\mathbf{x}, \mathbf{x}') = \exp\left(-\frac{\|\mathbf{x}-\mathbf{x}'\|^2}{2\ell^2}\right)$ (corresponds to infinite-dimensional ϕ)

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Kernelized Bayesian Regression: Setup

- Two equivalent expressions: primal (explicit features) and dual (only kernels)
- Feature-space model (Bayesian linear regression)

$$f(\mathbf{x}) = \phi(\mathbf{x})^T \mathbf{w}, \quad \mathbf{w} \sim \mathcal{N}(\mathbf{0}, \alpha^{-1}\mathbf{I}), \quad y = f(\mathbf{x}) + \varepsilon,\ \varepsilon \sim \mathcal{N}(0, \sigma^2)$$

- Training inputs $\mathbf{X} = [\mathbf{x}_1, \ldots, \mathbf{x}_N]$, feature matrix

$$\mathbf{\Phi} := \big[\phi(\mathbf{x}_1), \ldots, \phi(\mathbf{x}_N)\big] \in \mathbb{R}^{P \times N}$$

Primal Form (Feature Space)

- Data model in feature space

$$\mathbf{y} = \boldsymbol{\Phi}^T \mathbf{w} + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})$$

- Posterior over weights is Gaussian

$$p(\mathbf{w} \mid \mathbf{X}, \mathbf{y}) = \mathcal{N}(\boldsymbol{\mu}_w, \boldsymbol{\Sigma}_w)$$
$$\boldsymbol{\Sigma}_w = \left(\alpha \mathbf{I} + \sigma^{-2} \boldsymbol{\Phi} \boldsymbol{\Phi}^T\right)^{-1}$$
$$\boldsymbol{\mu}_w = \sigma^{-2} \boldsymbol{\Sigma}_w \boldsymbol{\Phi} \mathbf{y}$$

- Predict at $\mathbf{x}'$ with $\boldsymbol{\phi}' := \boldsymbol{\phi}(\mathbf{x}')$

$$f' \mid \cdot \sim \mathcal{N}(\mu', \sigma_{f'}^2)$$
$$\mu' = (\boldsymbol{\phi}')^T \boldsymbol{\mu}_w$$
$$\sigma_{f'}^2 = (\boldsymbol{\phi}')^T \boldsymbol{\Sigma}_w \boldsymbol{\phi}'$$

- Complexity dominated by inverting a $P \times P$ matrix (cost $\mathcal{O}(P^3)$)

From Primal to Dual

- Base kernel and Gram matrix

$$k(\mathbf{x}, \mathbf{x}') := \phi(\mathbf{x})^T \phi(\mathbf{x}'), \qquad \mathbf{K}_0 := \mathbf{\Phi}^T \mathbf{\Phi} \in \mathbb{R}^{N \times N}$$

- Kernel evaluations at a test point $\mathbf{x}'$

$$\mathbf{k}_0' := k(\mathbf{X}, \mathbf{x}') = \left[k(\mathbf{x}_1, \mathbf{x}'), \ldots, k(\mathbf{x}_N, \mathbf{x}')\right]^T \ (= \mathbf{\Phi}^T \phi'), \qquad k_0'' := k(\mathbf{x}', \mathbf{x}')$$

From Primal to Dual

- Dual predictive mean (two equivalent forms)

$$\mu' = (\phi')^T \boldsymbol{\mu}_w = (\mathbf{k}_0')^T \left(\mathbf{K}_0 + \alpha\sigma^2\mathbf{I}\right)^{-1} \mathbf{y}$$

- Dual predictive variance (latent)

$$\sigma_{f'}^2 = (\phi')^T \boldsymbol{\Sigma}_w \phi' = \frac{1}{\alpha} k_0'' - \frac{1}{\alpha} (\mathbf{k}_0')^T \left(\mathbf{K}_0 + \alpha\sigma^2\mathbf{I}\right)^{-1} \mathbf{k}_0'$$

- For noisy observation $y' = f' + \varepsilon$: $\sigma_{y'}^2 = \sigma_{f'}^2 + \sigma^2$
- Rule of thumb: use primal if $P \ll N$; use dual if $N \ll P$ or $P = \infty$

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Appendix: Multivariate Gaussian Distribution

Let $\mathbf{x} \in \mathbb{R}^{d_x}$ be a multivariate Gaussian distribution with the probability density function (pdf) is

$$\mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \Sigma_{xx}) = \frac{1}{(2\pi)^{d_x/2} \det(\Sigma_{xx})^{1/2}} \exp\left[-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^T \Sigma_{xx}^{-1}(\mathbf{x} - \boldsymbol{\mu})\right] \tag{6}$$

The mean and covariance matrix are

$$\mathbb{E}(\mathbf{x}) = \boldsymbol{\mu}, \quad Cov(\mathbf{x}) = \mathbb{E}\left[(\mathbf{x} - \mathbb{E}(\mathbf{x}))(\mathbf{x} - \mathbb{E}(\mathbf{x}))^T\right] = \Sigma_{xx}. \tag{7}$$

Appendix: Linear Gaussian System

If two random variables $\mathbf{x} \in \mathbb{R}^{d_x}$ and $\mathbf{y} \in \mathbb{R}^{d_y}$ have Gaussian distributions:

$$p(\mathbf{x}) \sim \mathcal{N}(\boldsymbol{\mu}_x, \Sigma_{xx}) \tag{8}$$

and

$$p(\mathbf{y}|\mathbf{x}) \sim \mathcal{N}(\mathbf{A}\mathbf{x} + \mathbf{b}, \Sigma_{yy}) \tag{9}$$

then the joint distribution is

$$p(\mathbf{x}, \mathbf{y}) \sim \mathcal{N}\left(\begin{bmatrix} \boldsymbol{\mu}_x \\ \mathbf{A}\boldsymbol{\mu}_x + \mathbf{b} \end{bmatrix}, \begin{bmatrix} \Sigma_{xx} & \Sigma_{xx}\mathbf{A}^T \\ \mathbf{A}\Sigma_{xx} & \mathbf{A}\Sigma_{xx}\mathbf{A}^T + \Sigma_{yy} \end{bmatrix}\right). \tag{10}$$

Here, $\mathbf{A}$ is a known constant matrix of size $d_y \times d_x$.

Appendix: Conditional Gaussian Distribution

If two random variables $\mathbf{x} \in \mathbb{R}^{d_x}$ and $\mathbf{y} \in \mathbb{R}^{d_y}$ are jointly Gaussian with the following joint distribution:

$$p(\mathbf{x}, \mathbf{y}) \sim \mathcal{N}\left(\begin{bmatrix} \boldsymbol{\mu}_x \\ \boldsymbol{\mu}_y \end{bmatrix}, \begin{bmatrix} \Sigma_{xx} & \Sigma_{xy} \\ \Sigma_{yx} & \Sigma_{yy} \end{bmatrix} \right), \tag{11}$$

then it is easy to derive the following conditional probabilities:

$$p(\mathbf{x}|\mathbf{y}) = \mathcal{N}(\boldsymbol{\mu}_{x|y}, \Sigma_{x|y}), \qquad p(\mathbf{y}|\mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}_{y|x}, \Sigma_{y|x}), \tag{12}$$

where

$$\boldsymbol{\mu}_{x|y} = \boldsymbol{\mu}_x + \Sigma_{xy}\Sigma_{yy}^{-1}(\mathbf{y} - \boldsymbol{\mu}_y), \qquad \boldsymbol{\mu}_{y|x} = \boldsymbol{\mu}_y + \Sigma_{yx}\Sigma_{xx}^{-1}(\mathbf{x} - \boldsymbol{\mu}_x), \tag{13}$$

and

$$\Sigma_{x|y} = \Sigma_{xx} - \Sigma_{xy}\Sigma_{yy}^{-1}\Sigma_{yx}, \qquad \Sigma_{y|x} = \Sigma_{yy} - \Sigma_{yx}\Sigma_{xx}^{-1}\Sigma_{xy}. \tag{14}$$

Appendix: Marginal Gaussian Distribution

Following the previous outcome of the joint Gaussian distribution $p(\mathbf{x}, \mathbf{y})$.

The marginal distributions out of it are still Gaussian, i.e.,

$$p(\mathbf{x}) = \int p(\mathbf{x}, \mathbf{y}) \mathrm{d}\boldsymbol{y} \sim \mathcal{N}\left(\boldsymbol{\mu}_x, \Sigma_{xx}\right), \tag{15}$$

and

$$p(\mathbf{y}) = \int p(\mathbf{x}, \boldsymbol{y}) \mathrm{d}\mathbf{x} \sim \mathcal{N}\left(\boldsymbol{\mu}_y, \Sigma_{yy}\right). \tag{16}$$

The End